



Final Year Project Proposal

**VisionNav: Mobile App for Visually
Impaired Mobility**

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1 Introduction

Independent mobility remains a major challenge for visually impaired individuals, especially within dynamic environments such as public transportation systems. According to World Health Organization estimates, approximately 2.2 billion people globally have vision impairment, with over 1 billion cases being preventable or unaddressed [1]. In Pakistan, around 1.76 million people are blind and over 24 million have some form of visual impairment [2]. Traditional mobility aids such as white canes and guide dogs provide physical assistance but lack contextual scene interpretation. They cannot detect distant obstacles, interpret bus route numbers, or provide spatial awareness beyond immediate contact [3].

Recent advancements in Artificial Intelligence (AI) and Computer Vision have enabled smartphones to perform real-time scene understanding. YOLO-series object detectors have achieved real-time performance with high accuracy on mobile devices [4, 7], while mobile OCR engines like Google ML Kit deliver excellent text recognition accuracy [5]. Smartphone-based solutions are particularly suitable in developing countries due to high mobile penetration and zero additional hardware cost.

VisionNav proposes a mobile-based AI assistive system that integrates real-time object detection, Optical Character Recognition (OCR), and CNN-based monocular depth estimation to provide contextual auditory guidance. The system converts visual environmental information into structured voice feedback, enhancing safety, independence, and confidence for visually impaired individuals when navigating public transportation.

2 Objective

- To develop a real-time YOLOv8-based object detection system to identify buses, pedestrians, vehicles, and obstacles in public transportation environments [4].
- To integrate Google ML Kit OCR to accurately recognize bus route numbers from display boards under varying environmental conditions [5].
- To utilize a lightweight CNN-based monocular depth estimation model to approximate object distances and generate proximity-based safety alerts [11, 12].
- To implement a Text-to-Speech module to convert detection and recognition results into real-time auditory guidance.
- To optimize AI models for efficient, low-latency mobile inference.
- To evaluate system performance using metrics such as mAP, Precision, Recall, IoU, F1-score, and OCR accuracy.

3 Problem Description

Visually impaired individuals face significant difficulty identifying buses at crowded stops and detecting obstacles beyond physical cane range. Existing GPS-based applications assist with routing but lack real-time environmental awareness and proximity information [3]. Many current assistive tools rely on internet connectivity or human volunteers (e.g., Be My Eyes), which are not always reliable in crowded public transport scenarios. There is no integrated mobile solution combining object detection, OCR-based bus recognition, and approximate distance estimation specifically tailored for public transport navigation in dynamic, low-resource settings such as those in Pakistan. This gap creates dependency, safety risks, and limited autonomy for visually impaired commuters.

4 Comparative Analysis

Comparative analysis evaluates existing solutions against the proposed VisionNav system, highlighting strengths, limitations, and the research gap.

Existing GPS-based tools (Google Maps, Lazarillo) provide route guidance but lack real-time scene understanding [13]. Volunteer-based platforms (Be My Eyes) offer human assistance yet require constant internet and are not real-time. Recent computer-vision systems focus on indoor navigation or general obstacles but exclude public-transport-specific OCR and localized datasets [14, 15].

VisionNav addresses these limitations through a unified, on-device architecture tailored for Pakistani urban environments.

Table 1: Comparative Analysis of VisionNav with Existing Solutions

Solution	GPS Navigation	Object Detection	Bus Route OCR	Monocular Depth	On-Device	Local Data
VisionNav (Proposed)	Lazarillo API (macro)	YOLOv8 (real-time)	Yes (ML Kit)	Yes (Mi-DaS)	Yes	Yes
Google Maps	Full routing	No	No	No	No	No
Lazarillo	Full routing	No	No	No	Partial	No
Be My Eyes	No	Human volunteer	No	No	No	No
Davanthapuram et al. (2021)	No	YOLO	No	Yes	No	No
Wang et al. YOLO-OD (2024)	No	YOLO	No	No	Partial	No

Research Gap: No prior mobile solution integrates Lazarillo-based macro-navigation with YOLOv8 object detection, ML Kit OCR for bus routes, and monocular depth estimation using a Pakistan-specific dataset.

5 Dataset Description

The dataset strategy combines benchmark pre-training with localized fine-tuning to ensure high accuracy in real-world Pakistani public transport scenarios.

5.1 COCO Dataset

The Microsoft COCO dataset provides over 330,000 images with 80 labeled categories (including bus, person, car, and truck), serving as a robust benchmark for pretraining [6].

5.2 Custom Local Dataset

A custom dataset of 800–1200 images will be collected from Islamabad and Rawalpindi bus stations, capturing local lighting, bus designs, and digital route boards. Annotations will be performed using Roboflow for domain-specific fine-tuning of YOLOv8.

6 Methodology

The methodology provides a structured framework for data acquisition, model development, system integration, and evaluation, ensuring reproducibility and real-world applicability.

6.1 GPS Navigation Module

To enable seamless navigation from home to bus stations, VisionNav integrates the Lazarillo API — a dedicated assistive platform designed specifically for visually impaired users. Lazarillo delivers turn-by-turn voice guidance, public transport information, and accessible routing. Upon reaching the bus stop (detected via user voice command or proximity), the system automatically activates the camera-based VisionNav module for object detection, OCR, and depth estimation [13].

6.2 Data Collection & Knowledge Integration

The COCO dataset will be used for object detection pretraining due to its 80 labeled object categories and benchmark reliability [6]. Additionally, a custom dataset of 800–1200 local bus stop images (collected in Islamabad/Rawalpindi area) will be prepared using annotation tools such as Roboflow.

6.3 AI Model

YOLOv8 is selected for its real-time detection speed and high accuracy [4]. Google ML Kit provides optimized on-device OCR functionality [5]. A lightweight CNN-based monocular depth estimation model (MiDaS small variant) will provide approximate distance measurements without any additional hardware [11, 12].

6.4 System Architecture

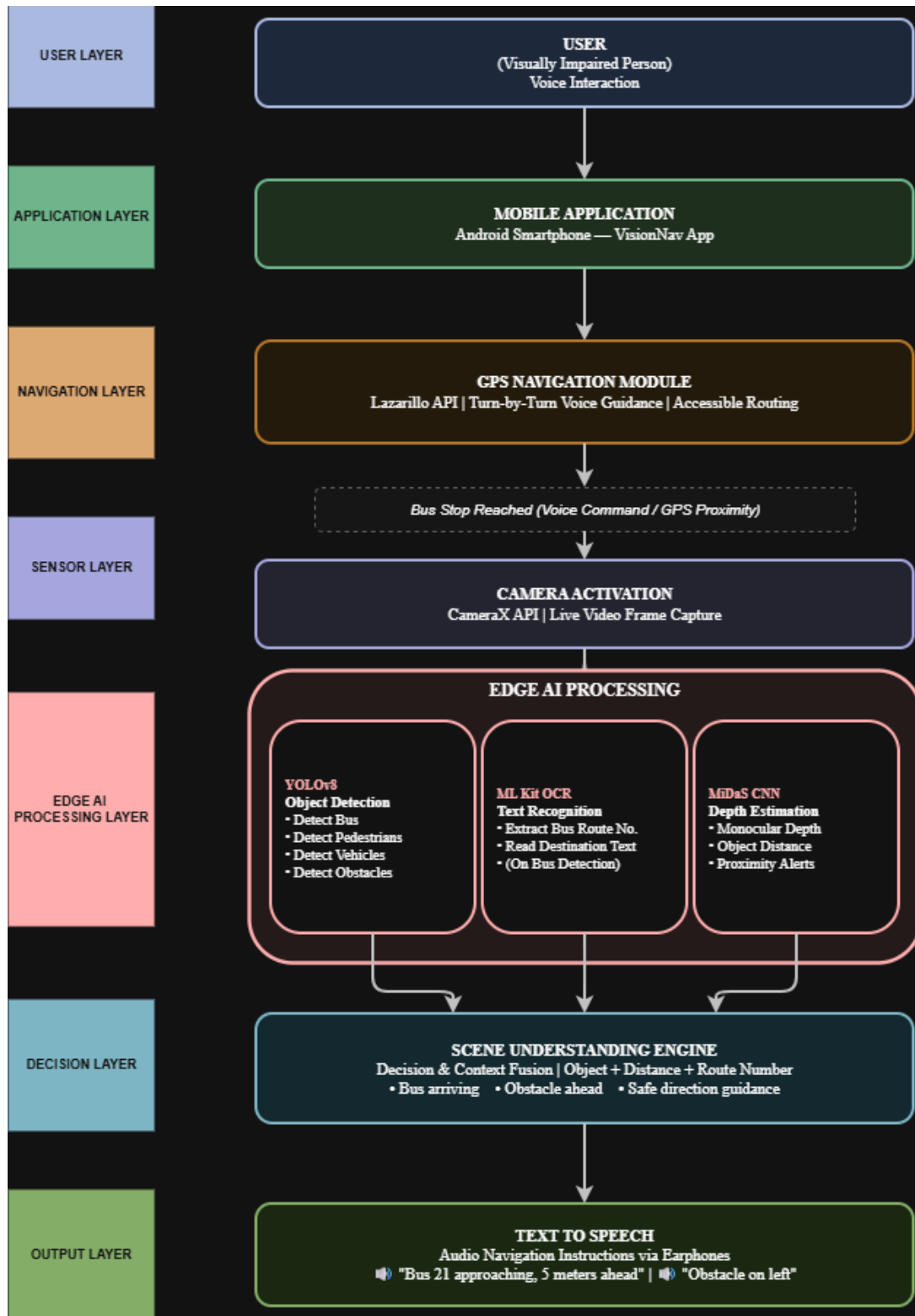


Figure 1: System Architecture of VisionNav using CNN-based Monocular Depth Estimation

Evaluation Metrics: Detection performance will be measured using mAP, Precision, Recall, IoU, and F1-score [7]. OCR performance will be measured using Character Accuracy Rate (CAR) and Word Recognition Accuracy [5].

7 Project Scope

The project scope clearly defines the boundaries to maintain focus and feasibility while delivering a complete, functional solution.

7.1 Inclusions

Real-time object detection, OCR for bus route numbers, lightweight CNN-based monocular depth estimation, GPS-based navigation via Lazarillo, and real-time Text-to-Speech auditory feedback. All core AI processing occurs on-device for privacy and offline functionality.

7.2 Exclusions

Wearable hardware, indoor mapping, government transport database integration, and smart traffic system connectivity.

7.3 Assumptions

Users will operate modern Android devices with sufficient processing capability in typical outdoor lighting conditions. Basic smartphone familiarity is assumed.

8 Feasibility Study

Feasibility analysis confirms that the proposed system is technically viable, resource-efficient, and aligned with current mobile AI capabilities.

8.1 Risks Involved

Low-light performance and latency are primary risks, mitigated by lightweight model optimization [8]. These challenges will be carefully addressed during model selection and training to ensure reliable performance in real-world public transport environments.

8.2 Resource Requirements

Hardware: Android device with sufficient processing power, laptop with minimum 8GB RAM.

Software:

1. Android Studio
2. Python 3.x
3. YOLOv8 Framework
4. Google ML Kit
5. Lazarillo API
6. OpenCV
7. Roboflow

9 Solution Application Areas

The solution targets real-world deployment scenarios where independent mobility for visually impaired citizens remains severely limited.

9.1 Target Domains & Beneficiaries

Primarily visually impaired individuals relying on public transportation systems in urban areas such as Islamabad and Rawalpindi [2].

9.2 Impact

The system will enhance independence, navigation safety, and confidence in urban mobility for visually impaired users [3]. By providing real-time contextual auditory information, VisionNav aligns with global sustainable development goals for inclusive mobility and can significantly reduce dependency on caregivers or volunteers.

10 Tools/Technology

The selected technology stack ensures cross-platform compatibility, real-time performance, and ease of development within an academic timeline.

10.1 Hardware Requirements

Android smartphone with sufficient processing capability. Laptop with minimum 8GB RAM.

10.2 Software Requirements

1. Android Studio
2. Visual Studio Code
3. Python 3.x
4. YOLOv8 [4]
5. Google ML Kit [5]
6. OpenCV [9]
7. Roboflow
8. Lazarillo API [13]

11 Expertise of the Team Members

The team combines complementary strengths in research, development, and documentation, ensuring balanced execution of all project phases.

M Abdullah Bin Abdul Aziz possesses strong expertise in AI model development, YOLOv8 training, system integration, backend optimization, and mobile deployment.

Maahin Khalid Malik brings specialized skills in research, documentation, dataset collection, comparative analysis, Android frontend development, and project coordination.

12 Milestones

The project follows an incremental development model with clearly defined phases, responsibilities, and deliverables to ensure timely completion and continuous progress tracking.

Table 2: Project Milestones and Timeline

Phase	Timeline	Responsible	Task
Topic Selection	Jan – Mid	Both	Research and dataset exploration
Proposal Sub- mission	Feb	Both	Documentation sub- mission
Dataset Collec- tion	March	Maahin	Data collection and annotation
Model Develop- ment	April – May	Abdullah	YOLOv8 training and evaluation [4]
OCR Integra- tion	June	Abdullah	ML Kit integration [5]
Frontend Devel- opment	June – July	Maahin	Android UI setup
Backend Opti- mization	July – August	Abdullah	Model deployment
System Integra- tion	September	Both	Combine modules
Testing & Evalu- ation	October	Both	Performance testing
Final Report	October	Maahin	Final documentation

12.1 Gantt Chart

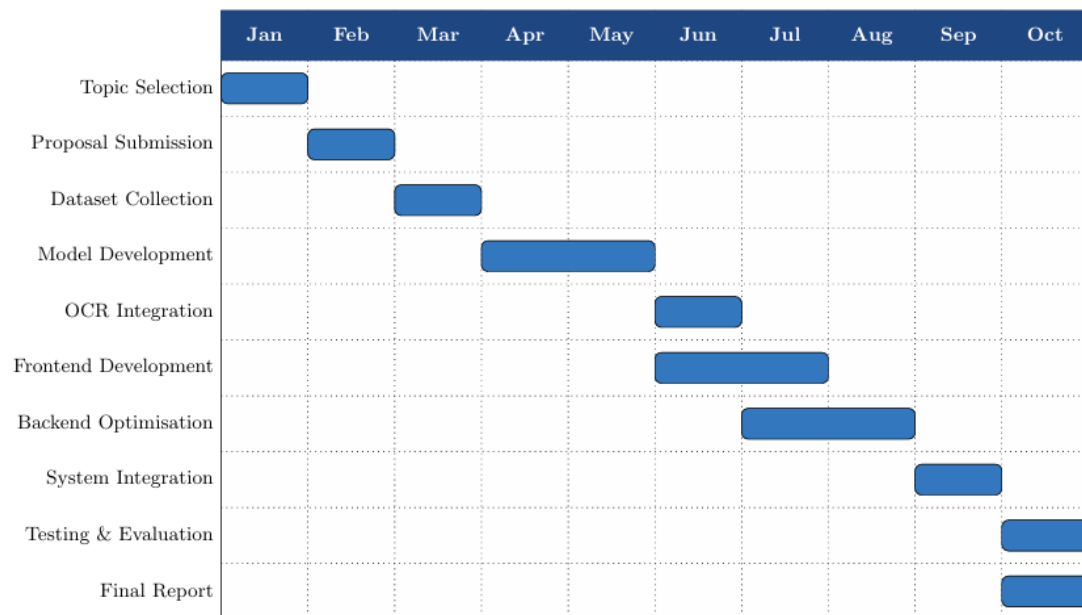


Figure 2: Project Timeline - VisionNav Development Schedule

The Gantt chart visually represents the project timeline, illustrating task dependencies and overlapping phases for efficient incremental integration.

13 References

References

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